

Data Cleaning Report

Project Title

Enterprise Data Preparation for Employee Attrition

Introduction

The purpose of this project is to prepare an employee HR dataset for further analysis and machine learning applications. Employee attrition is an important business problem because organizations need to understand why employees leave and identify patterns that may help improve employee retention.

Before using the data for analysis or prediction, it is necessary to clean and transform it properly. In this project, the dataset was checked for missing values, duplicate records, outliers, unnecessary columns, and inconsistent data types. The data was then prepared for further use through encoding, scaling, and feature engineering.

Dataset Overview

The HR Employee Attrition dataset contains information about employees, including their age, department, job role, income, work experience, job satisfaction, overtime, marital status, and attrition status.

The original dataset contained:

- **1,470 employee records**
- **35 columns**
- Both numerical and categorical variables
- **Attrition** as the main target variable

The dataset provides useful information for studying employee behaviour and identifying factors that may be associated with employee attrition.

Missing Value Analysis

The first step in data preparation was checking the dataset for missing values. Each column was examined to determine whether any information was unavailable.

The analysis showed that the dataset did not contain any missing values. Therefore, no records or features needed to be removed, and no missing-value imputation was required. This is beneficial because all employee records could be retained for further analysis.

Total Missing Values Found: 0

Duplicate Record Analysis

The dataset was also checked for duplicate employee records. Duplicate records can affect analysis by giving unnecessary importance to repeated observations and may produce misleading results.

No duplicate records were found in the dataset. Therefore, all 1,470 records were retained.

Total Duplicate Records Found: 0

Outlier Detection and Treatment

Outliers were examined using the Interquartile Range (IQR) method. Numerical features such as Age, MonthlyIncome, TotalWorkingYears, YearsAtCompany, DistanceFromHome, and promotion-related variables were checked.

Some HR variables naturally contain extreme values because employees have different salaries, experience levels, and years of service. Therefore, instead of deleting employee records, outliers were capped using the IQR boundaries where appropriate. This approach helps reduce the influence of extreme values while preserving valuable employee information.

This method was selected because removing records could result in the loss of important real-world employee cases.

Removal of Unnecessary Columns

The dataset contained some columns with only one constant value. Such columns do not provide useful information for analysis because they cannot help differentiate between employees.

The constant columns identified were removed from the processed dataset. In addition, the EmployeeNumber column was removed because it is only an employee identifier and does not provide meaningful information for attrition analysis.

Removing unnecessary columns helps reduce noise and improves the quality of the dataset.

Categorical Data Encoding

Several features in the dataset were categorical, including Department, BusinessTravel, Gender, JobRole, MaritalStatus, OverTime, and Attrition.

Machine learning algorithms generally require numerical input. Therefore, categorical variables were converted into numerical form using encoding techniques. Label encoding was used in the processed dataset, while One-Hot Encoding was included in the preprocessing pipeline for model-ready data preparation.

This transformation allows categorical employee information to be used effectively in analytical and machine learning processes.

Numerical Feature Scaling

The dataset contains numerical features with different ranges. For example, MonthlyIncome has much larger values compared with features such as JobLevel or WorkLifeBalance.

StandardScaler was used to standardize numerical features. This ensures that numerical variables are placed on a similar scale and prevents variables with larger values from having an unnecessary influence on later machine learning models.

The Attrition target variable was not scaled because it represents the outcome to be predicted.

Conclusion

The employee attrition dataset was successfully cleaned and prepared for further analysis. No missing values or duplicate records were found. Outliers were examined and treated carefully without unnecessarily removing employee records. Unnecessary and constant columns were removed, categorical variables were encoded, and numerical variables were scaled.

The final processed dataset is cleaner, more structured, and suitable for exploratory data analysis, machine learning, and employee attrition prediction.